Practice Quiz 11D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 11

Compare two proper infinite subsets.

1. Let $E = \{2n : n \in \mathbf{Z}^+\}$ and $M = \{3n : n \in \mathbf{Z}^+\}$. Define a bijection $F : E \to M$ and prove that it is one-to-one and onto.

   Solution Define $F(2n) = 3n$ for positive integers n. Every element of E has exactly one expression as $2n$, so this rule is well defined, and its value lies in M. If $F(2j) = F(2k)$, then $3j = 3k$, so $j = k$ and $2j = 2k$. Thus F is one-to-one. Given any $m \in M$, write $m = 3n$ with $n \in \mathbf{Z}^+$. Then $2n \in E$ and $F(2n) = m$, so F is onto. Hence $E \sim M$. ∎

Solutions prepared using GPT-6 in the Codex harness.